

App Encryption Declaration

VasBazaar — iOS application (bundle identifier `com.vasbazaar.app`)

Application name	VasBazaar
Bundle identifier	com.vasbazaar.app
Platform	iOS (Capacitor / WKWebView hybrid application)
Purpose of the app	Consumer fintech application for prepaid mobile and DTH recharges, utility bill payments (BBPS), wallet transactions and related financial services in India.
Declaration date	30 July 2026

1. Purpose of encryption in the app

Encryption is used solely to protect the confidentiality and integrity of the user's own data. It is a supporting security feature of a financial services application and is not the primary function of the app. Specifically, encryption protects:

- authentication credentials and session tokens exchanged with the VasBazaar backend;
- recharge and bill-payment request payloads (customer identifier, operator, amount);
- e-mandate / auto-pay setup requests;
- all API traffic between the app and the VasBazaar servers.

The app does not provide encryption as a service to the user, does not encrypt third-party data, and offers no key-management, secure-messaging or file-encryption features.

2. Encryption used

2.1 Transport security. All network communication with the VasBazaar backend uses HTTPS/TLS as provided by iOS (App Transport Security, `URLSession` / `WKWebView`). No custom TLS stack is bundled.

2.2 Application-layer payload encryption. Sensitive request payloads are additionally encrypted before transmission using a standard hybrid RSA + AES envelope scheme:

Payload cipher	AES-256 in CBC mode (256-bit key, 128-bit random IV, freshly generated per request)
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Key transport	RSA-OAEP with SHA-256 (MGF1-SHA-256). The RSA public key is issued by the VasBazaar backend and fetched over HTTPS; the app never holds a private key.
Random number source	<code>crypto.getRandomValues()</code> (Web Crypto, backed by the operating system CSPRNG)
Wire format	Base64 triplet: <code>RSA(AES key) : IV : AES(payload)</code>

2.3 Implementations used. Two interchangeable implementations of the same standard algorithms are present:

- **Web Crypto API** (`window.crypto.subtle`) — the primary path, provided by the operating system / WebKit;
- **node-forge** (open-source JavaScript library, BSD/GPL dual licence) — a fallback used only when the page is not running in a secure context and the Web Crypto API is therefore unavailable. It implements the same AES-CBC and RSA-OAEP algorithms in JavaScript.

Because the node-forge library is bundled with the app, the app implements standard encryption algorithms *in addition to* the encryption available in Apple's operating system. This is the reason the second option was selected in the App Store Connect encryption questionnaire.

3. Standards conformance

All algorithms used are published, internationally recognised standards. None are proprietary, and none have been modified:

- **AES** — FIPS PUB 197 (NIST); CBC mode per NIST SP 800-38A
- **RSA-OAEP** — PKCS #1 v2.2 / RFC 8017 (IETF)
- **SHA-256** — FIPS PUB 180-4 (NIST)
- **TLS** — RFC 8446 / RFC 5246 (IETF), as implemented by iOS

The app contains no proprietary or non-standard cryptography, no cryptanalytic functionality, and no ability for the user to supply or modify cryptographic algorithms.

4. Export compliance

The app is a mass-market consumer application distributed exclusively through the Apple App Store. Cryptography is limited to protecting the user's own data in transit and is ancillary to the app's primary function of processing recharges and bill payments.

On that basis the app is considered to fall under the mass-market provisions of Category 5, Part 2 of the U.S. Export Administration Regulations (ECCN 5D992.c). No CCATS classification or Encryption Registration Number has therefore been obtained. Consistent with this, the key `ITSAAppUsesNonExemptEncryption` is set to `false` in the application's `Info.plist`.

5. Distribution in France

The app is intended for use in India only and is not made available for distribution in France. No French encryption declaration has therefore been filed with ANSSI.

6. Statement

I confirm that the information provided above is accurate and complete to the best of my knowledge, and that the app uses only the standard, published encryption algorithms described in this document.

Name	_____
Title	_____
Company	_____
Signature	_____
Date	_____